MILITARY SPECIFICATION

WIRE ROPES, STEEL, BARE AND NYLON COVERED—— CORROSION-RESISTING, FLEXIBLE, PREFORMED, OCEANOGRAPHIC AND BATHYTHERMOGRAPHIC

1. SCOPE

1.1 Scope.- This specification covers requirements for flexible, preformed, corrosion-resisting steel wire rope, with and without nylon covering.

1.2 Classification.- Flexible, preformed, corrosion-resisting steel wire rope, covered by this specification, shall be of the following classes, as specified (see 6.2).

Class A - Without nylon covering.
Class B - With nylon covering.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
QQ-S-781 - Strapping, Flat; Steel.
QQ-S-790 - Steel Strapping, Round; (Bare and Zinc-Coated).
UU-P-271 - Paper, Wrapping, Waterproofed Kraft.

MILITARY
MIL-B-121 - Barrier Material, Greaseproofed, Flexible (Waterproofed).
JAN-T-781 - Terminal; Cable, Steel (for Swaging).

NAVY DEPARTMENT
General Specifications for Inspection of Material.

STANDARDS

FEDERAL
FED-STD-151 - Metals; Test Methods.

MILITARY
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

FSC 4010

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3. REQUIREMENTS

3.1 Wire material. - The wire used in the manufacture of the wire rope shall be of the corrosion-resisting type manufactured of induction furnace or electric arc furnace steel. The wire shall be suitable for manufacture into flexible, preformed wire rope, with or without nylon covering, which meets the requirements of this specification.

3.1.1 Chemical requirements. -

3.1.1.1 Chemical composition. - The check analysis made on the wire shall comply with the following chemical composition limits:

	Percent
(a) Carbon	0.15 maximum
(b) Manganese	2.00 maximum
(c) Silicon	1.00 maximum
(d) Phosphorous	0.045 maximum
(e) Sulphur	.030 maximum
(f) Chromium	17.0 to 20.0
(g) Nickel	8.0 to 12.0

3.1.1.2 An analysis of each heat or lot of steel from which the wire is made shall be furnished by the contractor. The ladle analysis shall meet the requirements of 3.1.1.1.

3.1.2 Mechanical properties. -

3.1.2.1 Tensile strength. - The wire shall have tensile strengths not less than the values shown in table I.

Table I - Minimum wire tensile strengths.

Wire diameter	Minimum tensile strength
Inch	Pounds per square inch (p.s.i.)
Up to but not including 0.011	280,000
0.011 up to but not including 0.021	260,000
.021 up to but not including .031	250,000
.031 up to but not including .051	240,000
.051 up to but not including .081	230,000
.081 up to and including 0.130	205,000

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3.1.2.1.1 In determining the tensile strengths p.s.i. of the wire tested, the actual diameter of the wire shall be used. Not less than one specimen of each diameter of wire from each strand shall be tested. Wire shall be considered of the same diameter when it is within the size tolerances shown in table II.

Table II - Wire diameter tolerances.

Wire diameter	Tolerances
Inch	Inch ±
Up to 0.029	0.00075
0.030 to 0.059	.0010
.060 to .099	.00125
.100 to .130	.0015

3.1.2.2 Wrapping.- The wire shall withstand the test specified in 4.2.6, without cracking or splitting.

3.2 Construction.-

3.2.1 Type of construction.- Wire rope shall be of the 7 by 7 construction. Wire rope of this construction shall consist of six strands of seven wires each laid around a core strand of seven wires. The wire shall be laid in accordance with the best commercial practice so as to develop a wire rope which will produce the greatest bending and wearing properties. The wire rope shall have a length of lay of not more than 8 times nor less than 6 times the wire rope diameter.

3.2.1.1 Preforming of wires and strands.- The individual wires and strands composing the wire rope shall be preformed into the exact helical position they will have in the finished wire rope, so that if the wire rope is cut or severed, there is no tendency for the normal diameter of the wire rope at the unseized ends to increase by more than the amount specified in table III.

Table III - Rope dimensional tolerances and physical properties.

Nominal diameter of wire rope	Tolerance on diameter (plus only)	Allowable increase in diameter at unseized end (max.)	Breaking strength of wire rope (min.)	Weight per 100 feet (approx.)
Inch	Inch	Inch	Pounds	Pounds
3/32	0.012	0.010	920	1.60
5/32	.016	.017	2200	4.50

3.2.1.2 Joining.- Splices in individual wires 0.014 inch in diameter and smaller may be twisted. Wires of larger diameter shall be brazed or welded. Splices in individual wires in any layer of a strand shall be not closer than 20 feet.

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3.2.1.3 Broken ends.- Protruding wires are considered as defects, and not more than one broken end will be accepted in 1000 feet of wire rope.

3.3 Class A - wire rope, without nylon covering.-

3.3.1 Dimensional tolerances and physical properties.- The dimensional tolerances and the physical properties for the diameters of uncoated wire rope shall be as specified in table III.

3.3.2 Endurance.- Class A, 5/32 inch diameter rope, shall resist the endurance test (see 4.3.3.3) at the load and for the number of reversals specified herein, without failure.

3.3.2.1 Strength after endurance test.- The breaking strength of the 5/32 inch diameter rope after the endurance test (see 4.3.3.4) shall be not less than the minimum values specified in table IV.

Table IV - Breaking strength after endurance test.

Rope diameter	Tension [1]	Number of reversals	Breaking strength (minimum)
Inch	Pounds		Pounds
5/32	30	75,000	1,300

[1] Tension is equal to half of weight (w) which includes idler sheave and hanger, as shown on figure 1.

3.3.3 Rope length.- Class A wire rope shall be furnished in reels of 3000 feet (for 3/32 inch diameter rope size) and 8000 and 20,000 feet (for 5/32 inch diameter rope size) continuous lengths, unless otherwise specified (see 6.2). All lengths shall be provided without joints of any kind, except as specified in 3.2.1.2.

3.3.4 Lubrication.- The wires of the class A rope shall be thoroughly coated with a suitable friction-preventive compound. The amount of compound adhering to the finished rope after the application shall be such that a white cloth, which has been rubbed once over a one-foot length of rope, will show a noticeable amount of the compound. The friction-preventive compound shall possess corrosion-resisting properties.

3.4 Class B - nylon covered wire rope.- The wire rope used in the manufacture of class B, wire rope shall be the same as specified in 3.2.1, 3.2.1.1, 3.2.1.2 and 3.2.1.3.

3.4.1 Nylon covering.- The nylon covering shall be DuPont Co. "ZYTEL 37X", or equal, 100 percent virgin nylon resin, having no reground material, pigmented or dyed black. The applied sheath shall be 1/32 inch thick, and shall be concentric with the wire rope. The outside diameter, after application of the covering, shall be 5/32 inch minimum.

3.4.1.1 The wire used in the fabrication of class B wire rope shall be the same as that specified for class A wire rope. The requirements of 3.1 are applicable.

3.4.2 Nylon finish.- Nylon finish shall be smooth, and free of waviness, scratches, pits, bubbles, grooves, cracks, checks, or inclusions. The covering shall be firmly composited with the wire rope.

3.4.3 Rope length.- Class B wire rope shall be furnished in reels of 250, 500, 1200, and 3000 feet, continuous lengths, unless otherwise specified (see 6.2). All lengths shall be provided without any joints of any kind, except as specified in 3.2.1.2.

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3.5 Workmanship.- All details of workmanship shall be in accordance with the best practice for high quality wire rope consistent with the requirements of this specification. The rope shall be securely laid and free from kinks, loose wires, loose strands, or other defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Wire.-

4.2.1 Lot.- Unless otherwise specified (see 6.2), a lot shall consist of all wire of the same class and size, submitted for acceptance inspection at the same time, and manufactured by the same process from steel of the same heat.

4.2.2 Chemical analysis samples.- From each lot, sufficient wire shall be taken to prepare a sample weighing not less than 2 ounces. The sample shall consist of fine chips of metal, free from oil, dirt, and other foreign matter which would interfere with a chemical analysis.

4.2.2.1 Chemical analysis.- Chemical analysis shall be made by standard wet chemical methods or by spectographic methods except that, in the event of a dispute, analysis by standard wet chemical methods shall be the basis for acceptance or rejection.

4.2.3 Sample coils for tensile strength and wrapping tests.- Samples for tensile strength and wrapping tests shall be selected by the inspector in accordance with table V. They shall be taken at random except that when inspection is made at the contractor's plant, some of the specimens shall be taken by the inspector near the first, last, and any intermediate coils of each lot, at the time the wire is being unreeled to form small coils for shipment.

Table V - Sample coils for tensile strength and wrapping.

Lot size in pounds	Number of sample coils	Maximum number of failures allowed
Under 1,000	2	0
1,000 to 3,000	3	0
3,001 to 10,000	5	0
Over 10,000	10	1

4.2.3.1 Test samples.- From each sample coil, a test sample long enough for test purposes to determine compliance with the requirements of this specification shall be taken by the inspector. Each test sample shall be identified by sample coil and lot number.

4.2.4 Examination.- Wire shall be visually and dimensionally examined for defects in accordance with the requirements of this specification not involving tests.

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4.2.5 Tests.- Unless otherwise specified herein, tests shall be conducted in accordance with Standard FED-STD-151.

4.2.6 Wrapping test.- The wire shall not crack, split or break when wrapped at least 8 turns in a closed helix (pitch approximately equal to the diameter of the wire) around a mandrel. For wire 0.162 inch in diameter and smaller the mandrel diameter shall be equal to the wire diameter.

4.2.7 Reworking and reinspection.- Rejected lots may be reworked by the manufacturer in order to eliminate material not conforming to this specification. After reworking, the material may again be submitted for inspection. Only one such resubmission shall be permitted for any lot of wire.

4.3 Finished rope.-

4.3.1 Lot.- A lot shall consist of wire rope of the same class, construction, and diameter produced continuously by one machine or one series of progressive processing machines.

4.3.2 Sampling.- When conducting the tests specified herein, with the exception of examination of product, (see 4.3.3.1), one sample, not less than 40 feet in length for the size specified in table IV and not less than 15 feet in length for other sizes, shall be taken, after any discard has been removed, from the head or starting end of the first manufacturing reel for each lot of rope. In addition to this sample, one sample shall be taken from each shipping reel of 20,000 feet or less, except that for shipping reels of 5,000 feet or less, one sample shall be taken from each 5,000 feet of rope from each lot.

4.3.3 Acceptance inspection.-

4.3.3.1 Examination.- All rope shall be examined for workmanship and finish. For this purpose the rope shall travel no faster than 100 feet per minute, and shall be stopped for closer examination when deemed necessary. This examination shall take place at the time the rope is wound on shipping reels. Any discard from the head end of the manufacturing reel shall be removed prior to winding on the shipping reels.

4.3.3.2 Breaking strengths.- Three specimens shall be prepared from each sample of rope selected. Terminals conforming to Specification JAN-T-781 shall be swaged to the ends of the specimens. The length of the rope test section, between terminals, shall be approximately 2 feet. The terminals shall be clamped in the jaws of the testing machine or suspended between fixtures attached to the heads. The breaking strength shall be determined by use of a tensile testing machine in compliance with the applicable requirement of method 211 of Standard FED-STD-151.

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4.3.3.3 Endurance. - One endurance test specimen shall be prepared from each sample of 5/32 inch diameter rope. The number of oscillations per minute, the total number of reversals, and the tension in the rope for the endurance test shall be as indicated on figure 1 and table IV for 5/32 inch diameter rope. The total travel of the rope in one direction shall be 13-1/2 inches. The test pulleys shall be made of steel and shall conform to the dimensions shown on figure 2.

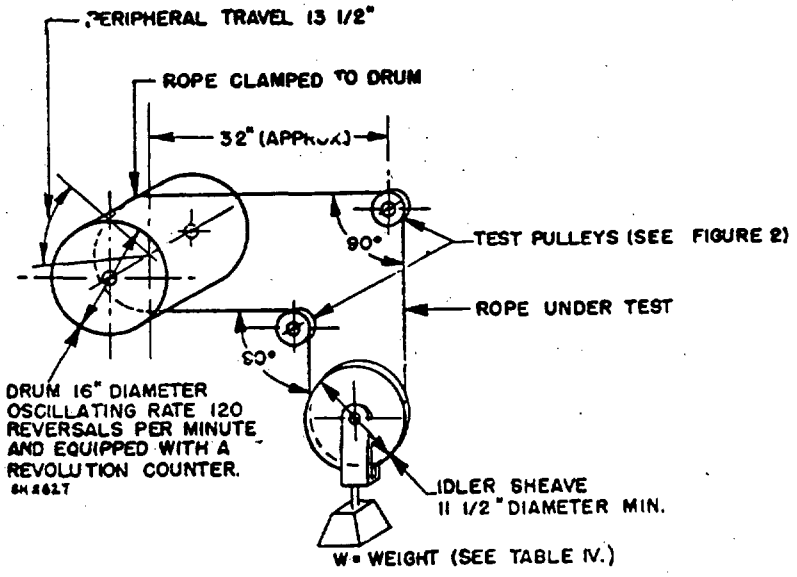
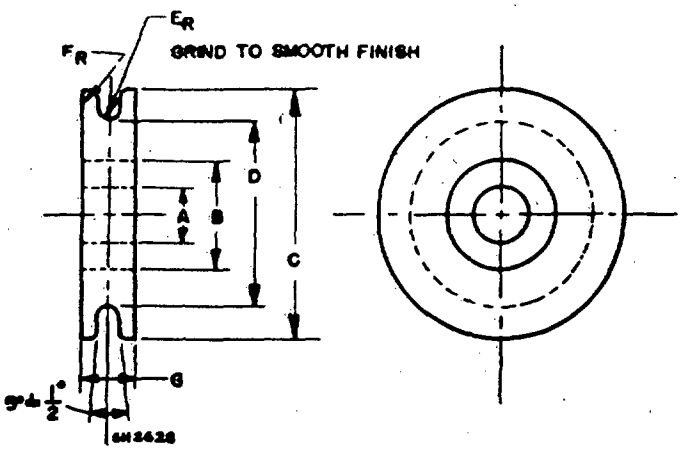


Figure 1 - Rope endurance testing machine (diagrammatic).

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NOTE: FIT PULLEYS WITH SUITABLE BALL OR ROLLER BEARINGS.
"B" SHOULD BE BORED AND GROUND TO LIGHT PRESS FIT FOR BEARINGS. "A" BORE OF BEARING AS RECEIVED.
MATERIAL: TOOL STEEL.
HEAT TREAT: HARDEN TO ROCKWELL C-60 MINIMUM.

DIMENSIONS OF STEEL PULLEYS USED IN ENDURANCE TESTS.

ROPE DIAMETER	PULLEY RATIO ↓	PULLEY DIMENSIONS				
		C ≙ 1/64	D +.005 −.000	E_R +.002 −.000	F_R	G ≙ 1/64
INCH		INCHES	INCHES	INCH	INCH	INCH
1/4	9.5	3 1/8	2.375	0.133	1/16	1/2

↓ RATIO OF PULLEY DIAMETER "D" TO NOMINAL ROPE DIAMETER.

Figure 2 - Endurance test pulley.

4.3.3.3.1 The application of lubricant to the endurance test sample, in addition to the lubricant applied during the manufacture of the rope, shall not be permitted either before or during the endurance test.

4.3.3.4 Breaking strength after endurance test. - The breaking strength, after the endurance test (see 4.3.3.3) shall be determined by the method specified for determining the breaking strength of rope (see 4.3.3.2) except that the specimen for this test shall be so selected as to determine the breaking strength of a portion of the rope that has been subjected to reverse bending by contact with a test pulley.

4.3.3.5 Determination of diameter. - The determination of the diameter of nylon covered wire rope shall be made at the free end of the rope on the shipping reel.

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4.3.3.5.1 When measuring the diameter there shall be no load on the rope. Beginning not less than 5 feet from the end of the rope, the diameter shall be measured at three places not less than 5 feet apart with a suitable device such as a machinists caliper square. The average of these diameters shall not exceed that specified in 3.4.1.

4.4 Rejection and retests.-

4.4.1 Rejection.- The failure of any specimen shall be cause for the rejection of the lot represented.

4.4.2 Retest.- The manufacturer may, at his own expense, in the presence of the inspector, sample each length of rope comprising a rejected lot and subject specimens from these samples to retest. The inspector may accept lengths of wire rope shown to conform to all requirements of this specification by this retest.

4.5 Inspection procedure.- For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 The wire rope shall be closely wound on wooden reels, full length, of one class and size to a reel. The inner and outer ends of the rope shall be securely attached to the reel.

5.2 Packaging.- Before starting to wind the wire rope on the reel, a layer of waterproof barrier-material, in accordance with Specification UU-P-271, and then a layer of greaseproof paper, in accordance with Specification MIL-B-121, shall be applied to the barrel of the reel and the inside of the flanges against which the wire rope will subsequently rest. After the wire rope is wound on the lined reel, a layer of greaseproof paper, in accordance with Specification MIL-B-121, then a layer of waterproof paper, in accordance with Specification UU-P-271, shall be applied and fastened by 3 or more wooden lagging strips and 2 metal straps. The metal strapping shall conform to Specification QQ-S-781 or QQ-S-790.

5.3 Packing.-

5.3.1 Level A.- When overseas packing is specified in the contract or order, each packaged reel shall be lagged with a layer of lumber in a manner that the waterproof barrier-material is not exposed to forces which may cause mechanical damage.

5.3.2 Level B.- For domestic packing, no additional packing is necessary.

5.4 Marking.- Shipments shall be marked in accordance with Standard MIL-STD-129. Item name for this material is "Wire Ropes, Steel, Bare and Nylon Covered—Corrosion-Resisting, Flexible, Preformed, Oceanographic and Bathythermographic".

6. NOTES

6.1 Intended use.- The wire rope covered by this specification is intended for oceanographic and bathythermographic use.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class required (see 1.2).
(c) Diameter required (see table III).
(d) Length required (see 3.3.3 and 3.4.3).
(e) Size of lot for wire (see 4.2.1).
(f) Selection of applicable level of packing (see 5.3).

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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